

JEE MAIN REVISION MASTER: STRUCTURE OF ATOM SYLLABUS REDUCED

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 219 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Quantum Numbers & Electronic Configuration	218 PYQs	Quantum Numbers
Very High	Bohr's Orbitals & Electronic Transitions	35 PYQs	Bohr Model
Very High	Heisenberg Uncertainty & De Broglie Wavelength	18 PYQs	Wave Mechanics

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Quantum Numbers & Electronic Configuration	$\text{Radial nodes} = n - l - 1$, $\text{Angular nodes} = l$, $\text{Total nodes} = n - 1$, $\mu_s = \sqrt{n(n+2)} \text{ BM}$
Bohr's Orbitals & Electronic Transitions	$r_n = 0.529 \frac{n^2}{Z} \text{ \AA}$, $E_n = -13.6 \frac{Z^2}{n^2} \text{ eV}$, $\bar{\nu} = \frac{1}{\lambda} = R_H Z^2 \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$
Heisenberg Uncertainty & De Broglie Wavelength	$\Delta x \cdot \Delta p \geq \frac{h}{4\pi} = \frac{\hbar}{2}$, $\Delta x \cdot \Delta v \geq \frac{h}{4\pi}$, $\lambda = \frac{h}{p} = \frac{h}{\sqrt{2mE}}$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Quantum Numbers & Electronic Configuration (Very High Yield — 218 PYQs)

Core Principle: Orbital shapes, quantum numbers (n, l, m_l, m_s), nodal surfaces, and anomalous configurations (Cr: $3d^5 4s^1$, Cu: $3d^{10} 4s^1$).

■ **Caution / JEE Pitfall:** Spin-only magnetic moment $\mu = \sqrt{n(n+2)}$ BM depends on n , which is the number of UNPAIRED electrons, not total electrons!

■ **Speed Trick:** Degeneracy of hydrogen atom at energy level n is n^2 (without spin) or $2n^2$ (including spin).

Representative Source PYQs:

- [2026 Evening Shift] Q#1 (MCQ): "The Bohr radius of a hydrogen like species is 70.53 pm . The species and the stationary state (n) are respectively (Given : Hydrogen atom $B_0 \dots$)" [Ans: N/A]
- [2026 Morning Shift] Q#1 (MCQ): "Arrange the following atomic orbitals of multi electron atoms in order of increasing energy. A. $n=3, l=2, m=+1$ B. $n=4, l=0, m=0$ C. $n=6, l=1, \dots$ " [Ans: N/A]
- [2026 Morning Shift] Q#1 (MCQ): "Which of the following is correct set of 4 quantum numbers of 19th electron in Chromium (Atomic number =24) in accordance with..." [Ans: N/A]

3.2 Bohr's Orbitals & Electronic Transitions (Very High Yield — 35 PYQs)

Core Principle: Electronic energy levels in hydrogenic species, wave number $\bar{\nu}$, and emission spectral series.

■ **Caution / JEE Pitfall:** Rydberg formula requires multiplication by Z^2 for single-electron ions like He^{+} ($Z=2$) and Li^{2+} ($Z=3$).

■ **Speed Trick:** Limiting line (series limit) of any spectral series occurs when $n_2 = \infty$ implies $\bar{\nu}(\text{limit}) = (R Z^2 / n_1^2)$.

Representative Source PYQs:

- [2026 Evening Shift] Q#1 (MCQ): "The Bohr radius of a hydrogen like species is 70.53 pm . The species and the stationary state (n) are respectively (Given : Hydrogen atom $B_0 \dots$)" [Ans: N/A]
- [2026 Evening Shift] Q#1 (MCQ): "Consider the following spectral lines for atomic hydrogen : A. First line of Paschen series B. Second line of Balmer series C. Third line of..." [Ans: C]
- [2026 Morning Shift] Q#2 (MCQ): "If shortest wavelength of hydrogen atom in Lyman series is x , then longest wavelength in Balmer series of He^{+} is:..." [Ans: A]

3.3 Heisenberg Uncertainty & De Broglie Wavelength (Very High Yield — 18 PYQs)

Core Principle: Limits of simultaneous precision in position and momentum, and dual wave nature of micro-particles.

■ **Caution / JEE Pitfall:** If uncertainty in position and momentum are equal ($\Delta x = \Delta p$), then $\Delta v = (1 / 2m)\sqrt{(h / \pi)}$.

■ **Speed Trick:** De Broglie wavelength of an electron accelerated through potential V volts: $\lambda \approx \frac{12.27}{\sqrt{V}} \text{ \AA}$.

Representative Source PYQs:

- [2026 Morning Shift] Q#2 (MCQ): "Identify the correct statements from the following : A. Heisenberg uncertainty principle is applicable to electrons. B. The size of $2p_x(x) \dots$ " [Ans: N/A]
- [2025 Evening Shift] Q#1 (MCQ): "Given below are two statements : Statement (I) : It is impossible to specify simultaneously with arbitrary precision, both the linear momentum..." [Ans: N/A]
- [2025 Morning Shift] Q#2 (MCQ): "If a_0 is denoted as the Bohr radius of hydrogen atom, then what is the de-Broglie wavelength (λ) of the electron present in the second or..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2026 (Online) 21st January Evening Shift (Q#1)

Consider the following spectral lines for atomic hydrogen : A. First line of Paschen series B. Second line of Balmer series C. Third line of Paschen series D. Fourth line of Brackett series The correct arrangement of the above lines in ascending order of energy is :

- (A) < C < A < B
- (B) < B < C < D
- (C) < A < C < B [Correct]
- (D) < D < B < A

Correct Answer: C

Solution Analysis: For hydrogen spectral lines (emission), $E = \frac{hc}{\lambda} \Rightarrow E \propto \frac{1}{\lambda}$ Using Rydberg formula (NCERT): $\frac{1}{\lambda} = R \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$ (n₂ > n₁) So photon energy is proportional to: $\Delta E = \frac{1}{n_1^2} - \frac{1}{n_2^2}$ Now identify each line: Paschen series : n₁=3 A (first line) : n₂=4 → 3 C (third line) : n₂=6 → 3 Balmer series : n₁=2 B (second line) : n₂=4 → 2 Brackett series : n₁=4 D (fourth line) : n₂=8 → 4 Compute Δ for each: D: $\frac{1}{8^2} - \frac{1}{4^2}$

Benchmark Problem 2: JEE Main 2026 (Online) 2nd April Evening Shift (Q#1)

The surface of sodium metal is irradiated with radiation of wavelength x nm. The kinetic energy of ejected electrons is 2.8×10^{-20} J. The work function of sodium is 2.3 eV. The value of x is _____ × 10² nm. (Nearest integer) (Given: $h = 6.6 \times 10^{-34}$ J s ; 1eV = 1.6×10^{-19} J ; c = 3.0×10^8 m s⁻¹)

Correct Answer: 5

Solution Analysis: Using Einstein's photoelectric equation, $h\nu = \phi + K_{\text{max}}$ Also, $h\nu = \frac{hc}{\lambda}$ So, $\frac{hc}{\lambda} = \phi + K_{\text{max}}$ Now convert the work function into joule: $\phi = 2.3 \text{ eV} = 2.3 \times 1.6 \times 10^{-19} \text{ J} = 3.68 \times 10^{-19} \text{ J}$ Given kinetic energy, $K_{\text{max}} = 2.8 \times 10^{-20} \text{ J}$ Hence total photon energy is $E = \phi + K_{\text{max}} = 3.68 \times 10^{-19} + 2.8 \times 10^{-20} = 3.96 \times 10^{-19} \text{ J}$ Now, $\lambda = \frac{hc}{E}$ Substitute the values: $\lambda = \frac{6.6 \times 10^{-34} \times 3.0 \times 10^8}{3.96 \times 10^{-19}} = 5 \times 10^2 \text{ nm}$